

The Structural Atlas of $\ker(Q)$

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Abstract

Five preceding papers in the K – Q operator framework computed one-loop observables in φ^4 theory, QED, and Yang-Mills theory, then extracted and stress-tested the ghost-activity theorem. Each paper found structure in the null space $\ker(Q)$ of the observable projection — but piecemeal, one type at a time. This paper maps $\ker(Q)$ systematically. We identify six structurally distinct types of elements in $\ker(Q)$, classified by their activity mode and loop-order stability: (1) scheme-dependent constants (inert, stable), (2) gauge ghost sectors (active/inert depending on algebra, stable), (3) unphysical polarizations (redistributive, stable), (4) anomalous currents (migratory — the key new result), (5) confined colored states (active non-perturbatively), and (6) topological sectors (latent). The central observation is that $\ker(Q)$ is not loop-order-stable: the chiral anomaly is a structural event in which an element of $\ker(Q)$ at tree level migrates to $\text{im}(Q)$ at one loop via the triangle diagram. This migration — not the anomaly coefficient itself — is the structural phenomenon. Types 1–3 are loop-stable; Type 4 is not; Types 5–6 are open. The classification provides the foundation for a representation-invariant decomposition theorem. All 24 structural results are machine-verified by Z3.

Keywords: null space, $\ker(Q)$, anomaly, chiral anomaly, confinement, topological sector, ghost, polarization, loop-order stability, structural atlas, Z3

1 Introduction

Five preceding papers [AtikPhi4, AtikQED, AtikYM, AtikGhost, AtikGauge] developed the K – Q operator framework for perturbative quantum field theory and found structure in $\ker(Q)$ — the null space of the observable projection — along the way. Each discovery was local: the φ^4 paper found scheme-dependent constants; the Yang-Mills paper found active ghosts; the gauge-dependence paper found representation-local attribution. But no paper asked the systematic question: **what is in $\ker(Q)$?**

This paper answers that question. We identify six structurally distinct types of elements living in $\ker(Q)$, classified by two properties:

1. **Activity mode:** how (or whether) the element couples into $\text{im}(Q)$ through $Q \circ K$.
2. **Loop-order stability:** whether the element remains in $\ker(Q)$ across loop orders.

The second property is new. The five preceding papers implicitly assumed that $\ker(Q)$ is static — that what is unobservable at tree level remains unobservable at all orders. This is true for Types

1–3 but false for Type 4. The chiral anomaly is a structural event in which an element of $\ker(Q)$ at tree level **migrates** to $\text{im}(Q)$ at one loop. This migration is the central observation of this paper.

The six types, in order of increasing structural complexity:

Type	Name	Activity	Loop-stable?
1	Scheme constants	Inert	Yes
2	Ghost sector	Active/inert	Yes
3	Unphysical polarizations	Redistributive	Yes
4	Anomalous currents	Migratory	No
5	Confined states	Active (non-pert.)	Open
6	Topological sectors	Latent	Open

Types 1–3 are established by the preceding papers. Type 4 is the key new analysis. Types 5–6 are structural predictions that the framework identifies but cannot compute perturbatively.

2 Type 1: Scheme-Dependent Constants

The simplest structure in $\ker(Q)$ consists of quantities whose values depend on the renormalization scheme: the tadpole integral A_0 , individual Passarino-Veltman scalar integrals $B_0(\rho)$, the Euler-Mascheroni constant γ_E , and logarithms of the renormalization scale $\ln(\mu^2)$.

These elements are in $\ker(Q)$ because they are not individually observable. In the $\overline{\text{MS}}$ scheme, $B_0(\rho)$ takes one value; in the on-shell scheme, another; in momentum subtraction, yet another. No experiment can determine which value is “correct” — because the question is not physical.

What **is** physical is the difference $B_0(\rho_1) - B_0(\rho_2)$, which is scheme-independent and lies in $\text{im}(Q)$. The observable projection Q annihilates the individual values and preserves only differences.

Activity: Inert. Scheme constants do not couple into $\text{im}(Q)$ through $Q \circ K$. They are passive cargo in $\ker(Q)$.

Loop-order stability: Yes. A scheme-dependent constant at one loop remains scheme-dependent at all orders. No mechanism promotes it to an observable.

Representation dependence: Yes — different schemes assign different values. But all schemes agree on $\text{im}(Q)$.

Z3-verified: A_0 and individual B_0 in $\ker(Q)$; scheme type inert; loop-order stable (4 results).

3 Type 2: Ghost Sector

Faddeev-Popov ghost fields are in $\ker(Q)$: they violate the spin-statistics theorem, carry no particle interpretation, and cannot appear as asymptotic states. The ghost-activity theorem [AtikGhost] established that the ghost sector $S_{\text{gh}} \subseteq \ker(Q)$ is active if and only if $f^{abc} \neq 0$.

The gauge-dependence paper [AtikGauge] refined this: ghost activity is an invariant of the covariant gauge family (ξ -independent) but not an invariant across all gauge-fixing schemes (in axial gauge, ghosts decouple). The physical content — that non-abelian structure forces the β -function sign — is representation-independent.

Activity: Active iff the gauge algebra is non-abelian. The algebraic switch is $f^{abc} \neq 0$.

Loop-order stability: Yes. Ghosts are unphysical at all loop orders. They never migrate to $\text{im}(Q)$ — they remain in $\ker(Q)$ while shaping $\text{im}(Q)$ through $Q \circ K$.

Representation dependence: Yes — FP-representation-specific. The diagrammatic mechanism changes across gauges; the observable content does not.

Z3-verified: ghost fields in $\ker(Q)$; ghost sector active; representation-dependent; loop-order stable (4 results).

4 Type 3: Unphysical Polarizations

In covariant gauges, the gauge boson propagator includes unphysical polarization states — longitudinal and scalar (timelike) modes. In Minkowski space, a massless spin-1 particle has two physical (transverse) polarizations. The covariant propagator,

$$D_{\mu\nu}(k) = \frac{g_{\mu\nu} - (1 - \xi)k_\mu k_\nu / k^2}{k^2}$$

propagates all four polarizations. The $(1 - \xi)$ term carries the longitudinal and scalar modes, which are in $\ker(Q)$: they cannot appear as asymptotic states.

These modes circulate in loops. The gluon loop contribution to β_0 depends on ξ — specifically, the longitudinal/scalar modes absorb the ξ dependence. The ghost loop is ξ -independent (established in [AtikGauge]). The total (gluon + ghost) is ξ -independent.

This reveals a new type of null-space activity: **redistributive**. The unphysical polarization sector does not shape a specific observable (like ghosts shape β_0). Instead, it absorbs representation dependence to keep the observable stable across the gauge family. It is the compensating mechanism that allows the total to be ξ -invariant despite individual contributions changing.

Activity: Redistributive — absorbs ξ variation to stabilize $\text{im}(Q)$.

Loop-order stability: Yes. Unphysical polarizations remain unphysical at all orders.

Representation dependence: Yes — the ξ -dependent contribution exists only in covariant gauges. In axial gauge, the modified propagator tensor structure absorbs both the ghost and polarization roles.

Z3-verified: longitudinal and scalar modes in $\ker(Q)$; redistributive type; loop-order stable (4 results).

5 Type 4: Anomalous Migration

This is the central new observation of this paper.

5.0.1 The classical picture

Consider the axial vector current $j_5^\mu = \bar{\psi}\gamma^\mu\gamma_5\psi$ in massless QED. Classically, this current is conserved:

$$\partial_\mu j_5^\mu = 0$$

This is the axial Ward identity. In K - Q language: the divergence of j_5 is zero at tree level, so it lies in $\ker(Q)$. It contributes nothing to any observable. It is, at tree level, as inert as the tadpole A_0 .

5.0.2 The one-loop picture

At one loop, the kernel K generates the triangle diagram (AVV): a fermion loop with one axial vertex and two vector vertices. This diagram is the Adler-Bell-Jackiw (ABJ) triangle [ABJ1969, Adler1969]. Its evaluation yields:

$$\partial_\mu j_5^\mu = \frac{e^2}{16\pi^2} F_{\mu\nu} \tilde{F}^{\mu\nu}$$

The right-hand side is nonzero. The divergence of the axial current is no longer zero. It is now an observable — it determines the decay rate $\pi^0 \rightarrow \gamma\gamma$, which is measured experimentally.

5.0.3 The structural event

In K - Q language, something unprecedented has happened: an element of $\ker(Q)$ at tree level has **migrated** to $\text{im}(Q)$ at one loop. The axial divergence was in $\ker(Q)$ (zero, unobservable) and is now in $\text{im}(Q)$ (nonzero, measurable). The anomaly is the structural event that moved it.

This means $\ker(Q)$ is **not loop-order-stable**. The boundary between $\ker(Q)$ and $\text{im}(Q)$ shifts when the loop order increases. The anomaly is the mechanism of migration.

5.0.4 Why this is different from Types 1–3

Types 1–3 are loop-stable: scheme constants remain scheme-dependent, ghosts remain unphysical, longitudinal modes remain non-asymptotic, at all loop orders. Their membership in $\ker(Q)$ is permanent.

Type 4 is not permanent. The element migrates. The classical symmetry (axial conservation) that placed it in $\ker(Q)$ is broken by the quantum correction. The anomaly is not a “failure” of the symmetry — it is a structural reclassification of the element from $\ker(Q)$ to $\text{im}(Q)$.

5.0.5 The anomaly coefficient is exact

The ABJ anomaly coefficient $e^2/(16\pi^2)$ is one-loop exact — it receives no corrections at higher loops (the Adler-Bardeen theorem [AdlerBardeen1969]). This means the migration event happens precisely at one loop. It is not a perturbative approximation; it is exact.

Activity: Migratory — moves from $\ker(Q)$ to $\text{im}(Q)$ across loop orders.

Loop-order stability: No. This is the defining property of Type 4.

Representation dependence: No — the anomaly is universal. It does not depend on regularization scheme or gauge-fixing choice (though different regulators may distribute it differently between axial and vector Ward identities).

Z3-verified: axial divergence in $\ker(Q)$ at tree level; migrates to $\text{im}(Q)$ at one loop; axial Ward identity holds classically, broken by anomaly (4 results).

6 Type 5: Confined Colored States

Quarks and gluons, as individual particles, are in $\ker(Q)$: no isolated color-charged state has ever been observed. This is confinement. Unlike Types 1–4, this is not a perturbative statement — it is an empirical fact about the non-perturbative theory.

Yet quarks and gluons **completely determine** the hadron spectrum, which is in $\text{im}(Q)$. Every meson, baryon, and glueball is a bound state of confined constituents. The elements of $\ker(Q)$ shape $\text{im}(Q)$ not through one-loop composition $Q \circ K$, but through non-perturbative bound-state formation.

The ghost-activity theorem is the perturbative shadow of this phenomenon. At one loop, ghosts (in $\ker(Q)$) shape β_0 (in $\text{im}(Q)$) through $Q \circ K$. Non-perturbatively, quarks and gluons (in $\ker(Q)$) shape the entire hadron spectrum (in $\text{im}(Q)$) through dynamics that the perturbative K cannot access.

Activity: Active — but the coupling route from $\ker(Q)$ to $\text{im}(Q)$ is non-perturbative.

Loop-order stability: Open. Quarks and gluons are in $\ker(Q)$ at all perturbative orders (they never appear as asymptotic states in perturbation theory). Whether “loop-order stability” even applies in the non-perturbative regime is an open structural question.

Representation dependence: Open. Confinement is a non-perturbative phenomenon; its representation dependence (if any) is not accessible to the current framework.

This type is stated as a structural prediction: the K – Q framework identifies it, classifies it, but cannot compute it. The perturbative boundary at Λ_{QCD} — where the one-loop $Q \circ K$ breaks down — is the edge of the framework’s domain, established in [AtikYM] and confirmed as gauge-invariant in [AtikGauge].

Z3-verified: isolated quarks and gluons in $\ker(Q)$; determine observables non-perturbatively; confined type active (4 results).

7 Type 6: Topological Sectors

The QCD vacuum has a topological structure parametrized by the vacuum angle θ . The θ parameter classifies degenerate vacua connected by instanton tunneling. The physical Lagrangian includes a topological term:

$$\mathcal{L}_\theta = \theta \frac{g^2}{32\pi^2} F_{\mu\nu}^a \tilde{F}^{a\mu\nu}$$

The angle θ is in $\ker(Q)$: it is not directly measurable. No experiment has ever determined its value. The strong CP problem is the experimental observation that θ is extremely small ($\theta < 10^{-10}$), despite no known symmetry requiring it to vanish.

If $\theta \neq 0$, it would produce CP violation in strong interactions — an $\text{im}(Q)$ effect. The neutron electric dipole moment would be proportional to θ . Empirically, this CP violation is absent, so θ remains inert.

We classify this as **latent**: structure in $\ker(Q)$ that could in principle migrate to $\text{im}(Q)$ (if $\theta \neq 0$) but empirically does not. The strong CP problem, in K – Q language, is: **why does this sector of $\ker(Q)$ remain inert?**

Instantons — the non-perturbative field configurations that tunnel between topological vacua — represent structure in $\ker(Q)$ invisible to the perturbative kernel K . They are a qualitatively different type of $\ker(Q)$ element: not a field, not a loop contribution, but a topological feature of the vacuum itself.

Gribov copies (multiple gauge-equivalent configurations satisfying the same gauge condition) add further structure: the space of Gribov copies is part of $\ker(Q)$ that the Faddeev-Popov procedure does not fully capture.

Activity: Latent — could migrate to $\text{im}(Q)$ but doesn't.

Loop-order stability: Open. Instantons are non-perturbative; they do not appear at any finite loop order.

Representation dependence: Open. Topological features are in principle gauge-invariant, but their description requires care.

Z3-verified: topological sectors latent; θ vacuum in $\ker(Q)$; θ is topological (4 results in the classification structure).

8 The Classification

The six types are summarized in the following table:

Type	Name	Status	Active?	Rep. dep.?	Loop-stable?
1	Scheme constants	Established	Inert	Yes	Yes
2	Ghost sector	Established	Iff $f^{abc} \neq 0$	Yes (FP)	Yes
3	Unphys. polarizations	New analysis	Redistributive	Yes (ξ)	Yes
4	Anomalous currents	New (key)	Migratory	No	No
5	Confined states	Prediction	Active (non-pert.)	Open	Open
6	Topological sectors	Prediction	Latent	Open	Open

8.0.1 The key structural property: loop-order stability

The property that most sharply separates the types is loop-order stability: whether an element of $\ker(Q)$ remains in $\ker(Q)$ when the loop order increases.

Types 1–3 are stable. Their elements are permanently in $\ker(Q)$. Scheme constants are always scheme-dependent. Ghosts are always unphysical. Longitudinal modes are always non-asymptotic. No amount of quantum correction changes their status.

Type 4 is unstable. The chiral anomaly moves an element from $\ker(Q)$ to $\text{im}(Q)$ in exactly one step (tree to one loop). The migration is exact (Adler-Bardeen theorem) and universal (scheme- and gauge-independent).

Types 5–6 are open. Confinement and topological structure are non-perturbative; the question of loop-order stability may not even apply in its perturbative form.

8.0.2 Activity modes

The classification identifies four distinct activity modes:

1. **Inert**: present in $\ker(Q)$, no coupling to $\text{im}(Q)$. (Type 1)
2. **Active**: in $\ker(Q)$ but shapes $\text{im}(Q)$ through $Q \circ K$. (Types 2, 5)
3. **Redistributive**: absorbs representation dependence to stabilize $\text{im}(Q)$. (Type 3)
4. **Migratory**: moves from $\ker(Q)$ to $\text{im}(Q)$ across loop orders. (Type 4)
5. **Latent**: could migrate but empirically doesn't. (Type 6)

These modes are not ad hoc labels — each corresponds to a distinct structural mechanism verified in the preceding papers or identified in this one.

Z3-verified: all 24 results across 6 structures.

9 Implications

9.0.1 $\ker(Q)$ is not homogeneous

The standard QFT treatment of renormalization treats $\ker(Q)$ as waste — divergent intermediates to be subtracted. The K – Q framework has progressively revealed that $\ker(Q)$ is structured: it contains active sectors, redistributive sectors, and now migratory sectors. The null space is not a featureless void; it is an organized space with internal distinctions that matter for observables.

9.0.2 Anomalies as boundary events

The most important new observation is that anomalies are **boundary events of $\ker(Q)$** : structural transitions in which elements cross from $\ker(Q)$ to $\text{im}(Q)$. This reinterpretation does not change the physics of the chiral anomaly — the triangle diagram, the ABJ coefficient, the $\pi^0 \rightarrow \gamma\gamma$ prediction are all unchanged. What changes is the structural context: the anomaly is not

merely “a classical symmetry broken by quantum effects” but “a migration event at the boundary of $\ker(Q)$.”

This reinterpretation unifies the anomaly with the other $\ker(Q)$ phenomena. Ghosts are active $\ker$ elements. Scheme constants are inert $\ker$ elements. Anomalous currents are **transient** $\ker$ elements — they pass through $\ker(Q)$ on their way to $\text{im}(Q)$. The classification makes these three cases instances of a single structural framework.

9.0.3 The atlas as foundation

The six-type classification provides the foundation for the representation-invariant decomposition theorem identified in [AtikGauge]: different gauge-fixing schemes correspond to different factorizations of $Q \circ K$ with the same image and different kernel realizations. The atlas specifies **what** is being redistributed (Types 1–3), **what** migrates (Type 4), and **what** lies beyond the perturbative domain (Types 5–6).

9.0.4 What this atlas does and does not claim

This atlas claims: $\ker(Q)$ contains at least six structurally distinct types of elements, classified by activity mode and loop-order stability, with the anomalous migration type being the key new identification.

This atlas does not claim: that these six types are exhaustive (there may be others), that Types 5–6 are computable in the current framework (they are not), or that the classification extends straightforwardly beyond one loop (the Adler-Bardeen theorem gives exactness for Type 4; the other types require separate analysis at higher orders).

The atlas is a map, not the territory. But it is the first systematic map of $\ker(Q)$, and it identifies the most important structural property — loop-order stability — that was invisible before the anomalous migration type was recognized.

9.0.4 References

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